

Bowen Qiu

PhD Student | The University of Hong Kong

Department of Electrical and Computer Engineering

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PhD researcher in MRI and neuroimaging, with experience in ultra-low-field brain imaging, adolescent cognition, fetal brain analysis, and biomedical software.

Education

The University of Hong Kong	2025 - present
PhD Student, Electrical and Computer Engineering	
Supervisor: Prof. Ed X. Wu	
Zhejiang University	2021 - 2025
B.E. in Biomedical Engineering; Minor in Finance	
Supervisors: Mingyang Li and Dan Wu. GPA: 3.99/4.00; ranked 1st out of 133 students	
Oxford Prospects and Global Development Institute	Aug. 2024
Two-week Interdisciplinary Academic Programme	
Oxford, UK. Completed 46 hours of lectures, workshops, guest lectures, and independent study; received an A- (Excellent) grade	

Research Experience

Ultra-low-field MRI and brain age	2025 - present
The University of Hong Kong	
- Investigating brain-age prediction with 0.055T ultra-low-field MRI.	
- Presented this work as a digital poster at ISMRM 2026.	
Fetal brain MRI segmentation analysis	Jul. - Nov. 2024
Boston Children's Hospital / Harvard Medical School	
- Analyzed regional error patterns in automated cortical plate segmentation of fetal brain MRI and compared pretrained U-Net and fuzzy U-Net models.	
- Presented the analysis as a digital poster at OHBM 2025.	
Adolescent intelligence and multimodal MRI	2023 - 2025
Zhejiang University	
- Used ABCD and Allen Brain datasets to examine crystallized and fluid intelligence through MRI, spatial correlation, and gene enrichment analyses.	
- First-author work was published in Translational Psychiatry (2025) and presented at ISMRM 2025.	

Zhejiang University

- Led a National University Student Innovation and Entrepreneurship Development Program project on adolescent brain imaging and cognition.

- Coauthored a 2026 medRxiv preprint on longitudinal central adiposity, cortical changes, and cognition.

Selected Publications

Journal articles

Neural correlates differ between crystallized and fluid intelligence in adolescents

Bowen Qiu, Rui Qian, Baorong Gu, Zichao Li, Zhifan Chen, Xinyi Xu, Huaijin Gao, Yiwei Chen, Ruoke Zhao, Ruike Chen, Yuqi Zhang, Lingxuan Zhang, Zhiyong Zhao, Mingyang Li, Dan Wu

Translational Psychiatry, 15, 246 (2025).

[Paper](#) | [PDF](#) | [DOI](#)

Distinct and Shared Molecular Mechanisms Underlie Morphological-Functional Overcoupling and Undercoupling in Major Depressive Disorder

Huaijin Gao, Yihan Ma, Rui Qian, Tao Jin, Baorong Gu, **Bowen Qiu**, Shaoyong Ye, Manli Huang, Dan Wu, Zhiyong Zhao

Biological Psychiatry, 2026 | In press; available online 25 June 2026.

[Paper](#) | [DOI](#)

Preprints

Preprints have not been certified by peer review.

Longitudinal Central Adiposity Accumulation is Associated with Cortical Alteration and Impaired Cognitive Function in Adolescents

Lingxuan Zhang, **Bowen Qiu**, Zhifan Chen, Xinyi Xu, Ruoke Zhao, Yiwei Chen, Chenglin Ning, Ruike Chen, Mingyang Li, Dan Wang, Junfen Fu, Dan Wu

medRxiv, 2026 | Preprint, not peer reviewed.

[Preprint](#) | [PDF](#)

Multi-Scale Characterization of Distance-dependent Functional Dysconnections in Major Depressive Disorder Across Two Asian Cohorts: A Genomic, Neurochemical, and Cellular Perspective

Rui Qian, Huaijin Gao, **Bowen Qiu**, Zichao Li, Baorong Gu, Tongmei Ye, the DIRECT Consortium, Dan Wu, Zhiyong Zhao

medRxiv, 2024 | Preprint, not peer reviewed.

[Preprint](#)

Conference Abstracts and Posters

Human Brain Age Prediction using 0.055T Ultra-low-field MRI

Bowen Qiu, Vick Lau, Xuehong Lin, Junhao Zhang, Xiang Li, Gilberto K. K. Leung, Yujiao Zhao, Alex T. L. Leong, Ed X. Wu

ISMRM, 2026 | Digital poster | Program 562-03-010.

[Program](#)

Regional Analysis of Error Patterns from Automatic Cortical Plate Segmentation in Fetal Brain MRI

Bowen Qiu, Andrea Gondova, Sungmin You, Milton O. Candela-Leal, Hanuman Verma, Seungyoon Jeong, P. Ellen Grant, Kiho Im

OHBM, 2025 | Digital poster.

Neural correlates differ between crystallized and fluid intelligence in adolescents: A Multi-modal MRI Investigation

Bowen Qiu, Rui Qian, Baorong Gu, Zichao Li, Zhifan Chen, Xinyi Xu, Yiwei Chen, Ruoke Zhao, Ruike Chen, Yuqi Zhang, Zhiyong Zhao, Mingyang Li, Dan Wu

ISMRM, 2025 | Digital poster | Abstract 4863.

[Abstract](#)

Distance-Dependent Functional Dysconnectivity and Neurotransmitter Correlates in Major Depressive Disorder

Rui Qian, Huaijin Gao, **Bowen Qiu**, Baorong Gu, Zichao Li, Dan Wu, Zhiyong Zhao

ISMRM, 2025 | Oral presentation | Abstract 0504.

[Abstract](#)

Cerebrovascular Reactivity Development in Infants during the First Two Years of Life Using Resting-State BOLD Functional MRI

Baorong Gu, Rui Qian, Zilong Ye, **Bowen Qiu**, Zichao Li, Mingyang Li, Ruoke Zhao, Yuqi Zhang, Yutian Wang, Peiying Liu, Hanzhang Lu, Dan Wu, Zixuan Lin

ISMRM, 2025 | Digital poster | Abstract 2566.

[Abstract](#)

Selected Honors and Awards

ISMRM Trainee (Educational) Stipend	2026
Hong Kong PhD Fellowship Scheme (HKPFS)	2025
HKU Presidential PhD Scholarship	2025
Chu Ko Chen Scholarship, Zhejiang University	2024
Top-10 Outstanding Students, Zhejiang University	2024
National Scholarship (three awards)	2022, 2023, 2024

Engineering Projects

Intestinal Health Assistant

2023 - 2024

Led a seven-member team developing a WeChat mini program for recording stool characteristics and supporting health monitoring; contributed data upload, download, and local storage functions.

Brain tumor image classification

Apr. 2024

Adapted ResNet50 with transfer learning and image preprocessing for a medical AI course project.

Patent and Software Copyrights

- Smart toilet with seat for health monitoring. Chinese patent application No. 202410326127.6.
- Intestinal Health Assistant, a WeChat mini program. Software copyright registration No. 2023SR1496402.
- Atlas2vec, a toolbox for mapping brain sub-atlases. Software copyright registration No. 2024SR1598299.

Leadership and Teaching

Student Chairman, Zhejiang University Wind Orchestra

2023 - 2024

Organized recruitment and three concerts; helped lead the ensemble to first-prize awards in Zhejiang Province and the Yangtze River Delta region.

Peer Tutor and Calculus Teaching Assistant, Zhejiang University

2022 - 2023

Led academic tutoring sessions, taught calculus, graded work, and supported younger students outside class.

Student social practice

2022 - 2023

Led brain bank outreach and alumni-visit activities; participated in the Snow Dragon ship scientific practice programme.

Additional Academic Activities

- CUHK Faculty of Engineering PhD Summer Workshop (Jul. 2024).
- Research team exchange visit, The Hong Kong Polytechnic University (Jan. 2024).

Selected Competitions

- Third Prize, National College Biomedical Engineering Innovation Design Competition (2024).
- First Prize, National Undergraduate Mathematics Competition, non-mathematics major (2023).
- First Prize, Zhejiang Provincial College Student Physics Innovation Competition, theoretical category (2023).
- Bronze Medal, Zhejiang University Internet+ Innovation and Entrepreneurship Competition (2023).

Additional Information

Programming: Python, MATLAB, R, JavaScript, and C.

Interests: Trumpet, swimming, and table tennis.